Memorization and generalization in LLMs. Whether models memorize training data or generalize to unseen data has been extensively studied in deep learning (e.g., Zhang et al., 2021) and specifically in LLMs (Tänzer et al., 2022; Carlini et al., 2023; Antoniades et al., 2024), but not many studies have observed this question through the lens of model internals. Among those that do, Bansal et al. (2022) attempt to predict this trade-off by observing the diversity of internal activations; Dankers & Titov (2024) show memorization in language classification tasks is not local to specific layers, and Varma et al. (2023) explain grokking using memorizing and generalizing circuits. We use this lens to observe how model internals operate in arithmetic reasoning—a task that could theoretically be solved either through extensive memorization or by learning a robust algorithm. Concurrent work (jylin et al., 2024) has shown that a LM trained to predict legal board game moves (Li et al., 2022) does so by implementing many heuristics. While heuristics would suffice to robustly predict legal moves in a board game setting, we find that the extent to which LLMs rely on heuristics is greater than prior work suggests: sets of heuristics are used to accomplish even generic tasks like arithmetic, where no heuristic is likely to generalize to all possible results.

Arithmetic reasoning interpretability. Recent studies on how LMs process arithmetic prompts (Stolfo et al., 2023; Zhang et al., 2024) reveal the general structure of arithmetic circuits, but do not fully explain *how* they combine operand information to produce correct answers. Our research bridges this gap by revealing the mechanism used for promoting the correct answer. Some studies show the emergence of mathematical algorithms for modular addition (Nanda et al., 2023; Zhong et al., 2024; Ding et al., 2024) and binary arithmetic (Maltoni & Ferrara, 2023) in simple, specialized toy LMs, but it is unclear if these findings extend to larger, general-purpose LMs or other operators. In pre-trained LLMs, Zhou et al. (2024) found that Fourier space features are used for addition. However, we claim this is only a partial view, as many additional types of features and heuristics relying on these features are involved in calculating answers across arithmetic operations. In this work, we give a wide view of these heuristics and how they combine to generate arithmetic answers.

7 Conclusions

Do LLMs rely on a robust algorithm or on memorization to solve arithmetic tasks? Our analysis suggests that the mechanism behind the arithmetic abilities of LLMs is somewhere in the middle: LLMs implement a *bag of heuristics*—a combination of many memorized rules—to perform arithmetic reasoning. To reach this conclusion, we performed a set of causal analysis experiments to locate a circuit, i.e., a subset of model components, responsible for arithmetic calculations. We examined the circuit at the level of individual neurons and pinpointed the arithmetic calculations to a sparse set of MLP neurons. We showed that each neuron acts as a memorized heuristic, activating for a specific pattern of inputs, and that the combination of many such neurons is required to correctly answer the prompts. In addition, we found that this mechanism gradually evolves over the course of training, emerging steadily rather than appearing abruptly or replacing other mechanisms.

Our results, showing LLMs' reliance on the bag of heuristics, suggest that improving LLMs' mathematical abilities may require fundamental changes to training and architectures, rather than post-hoc techniques like activation steering (Subramani et al., 2022; Turner et al., 2023). Additionally, the evolution of this mechanism across training indicates that models learn these heuristics early and reinforce them over time, potentially overfitting to early simple strategies; it is unclear if regularization can improve this, and this is a possible avenue for future research.

8 Limitations and discussion

Interpretability work is often fundamentally limited by human biases. As researchers, we often impose human abstractions onto models, whereas the goal of interpretability is to understand the abstractions that models learn and apply in a way that we can understand. Our work is also subject to this limitation, namely with respect to the definition of heuristic types: We define heuristic abstractions based on our human-identifiable definitions. A possible improvement would be to develop a method to identify these abstractions without human bias. Another important detail is that our analysis focuses on LLMs that combine digits in tokenization. That is, every token can contain more than one digit. The robust algorithms used by humans depend on our ability to separate larger numbers to single digits. Thus, a similar analysis might lead to different conclusions for models that perform single-digit tokenization.

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